

Tool/Feature	GCP Native Services
Visualisation report for cost and utilisation reports	Cost management provides visualisation reports on utilisation, filtered by labels. Data Studio is used to build custom dashboards.
Integration with third party and native tools	Billing API and data collection APIs (StackDrive, Resource Manager API) are used for integration with native and third party cost management tools.
Predictive analysis for cost advisory	Intelligent recommendations optimise cost and usage based on patterns of resource usage. Resource hierarchy is used for fine grained resource management for cost allocation.
Infra optimisation	Quota limits are used for proactive control of the spend rate on resources including apps and infrastructure.
Alert on usage patterns	Alerts with automated budget actions are used to throttle resources and cap costs.
Rule driven event handling	Alerts can be sent through events (SMS, email) using programmatic budget notification.
DevOps pipeline (native) integration	Can be integrated with Cloud Build, Cloud Pub/Sub (event/alerts) and cloud functions to invoke cost reporting, billing and notification.
Service now integration for provisioning activities (e.g., Datalink)	Cloud Connector app and StackDriver API can be used to integrate with ServiceNow.

Table 1: Google's cloud management services

Cloud service platforms like MS Azure, AWS and GCP provide native cloud management facilities like budgets, alerts, cost advisories, and workflow management.

Native cloud management vs cloud agnostic solutions

Gartner's guidance framework for selecting cloud management platforms and tools helps with cross-platform solutions for a multi-cloud or hybrid cloud approach, as well as with a platform-specific approach for native cloud adoption.

Infrastructure as Code (IaC), as well as cloud automation for provisioning and virtualisation needs, is very helpful in accelerating the building and deployment of cloud services. Preparing provisioning scripts for each cloud is inefficient, and IaC helps to automate these across different cloud environments with ease.

Though cloud native facilities like CloudFormation in AWS, ARM templates in Azure and CDM templates in GCP, enable IaC, they are mostly native to the cloud platform and don't have the agility and flexibility for multi-cloud management. On the other hand, services like Google Anthos, HashiCorp Terraform, and Red Hat Ansible can help with multi-cloud management, developing zero-touch deployment scripts and provisioning templates for setting up cloud towers.

Google Anthos, HashiCorp Terraform and Red Hat CloudForms are leading this space for creating, changing and provisioning infrastructure for any cloud platform including service management, security and compliance, and optimisation. Red Hat CloudForms has five integral components, i.e.,

CloudForms management engine appliance, CloudForms management engine server, virtual management database, CloudForms management engine console and smart proxy.

Challenges in multi-cloud management

A multi-cloud approach does have complexity and compatibility issues, as well as challenges with respect to security and monitoring.

1. **High complexity:** One of the hidden costs of multi-cloud computing is high complexity. Every cloud service provider has its own set of conventions, commands and ways of performing things. As an example, AWS has three load balancers while Google Cloud has six, and each performs different aspects of load balancing. Some work at Layer 7 and some at Layer 4, and deciding how to effectively balance load and keep applications secure is one of the complex issues to solve. Doing this across multiple clouds also increases complexity.
2. **Varied service capabilities across clouds:** Different clouds have different capabilities even with respect to offerings like computing, storage and networking. For example, in theory, Kubernetes should be managed in the same manner across all clouds. In practice, each major cloud provider has key differences in how it manages Kubernetes, the monitoring and security tools for these managed offerings, and performance levels. So even when a service is, on the face of it, exactly the same and built on the same core technology, there may be key differences underneath that can have a real impact on performance, resilience, and how you architect an application.